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SUPERSEDING
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29 October 1963

MILITARY SPECIFICATION

STAND, TEST, AIRCRAFT HYDRAULIC SYSTEM, ELECTRIC MOTOR DRIVEN, TYPE MK-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electro-motor-driven, trailer mounted, 27 gpm at 3,000 psi or 15 gpm at 5,000 psi, aircraft hydraulic system test stand, designated type MK-1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GG-G-75	Gage, Pressure and Vacuum, Dial Indicating (for Air, Steam, Oil Water, Ammonia, Chloro-Flouro Hydrocarbon Gases and Compressed Gases).
PPP-T-60	Tape; Packaging, Waterproof.

MILITARY

MIL-C-104	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	Preservation, Methods of.
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

AMSC N/A

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DOD-D-1000	Drawings, Engineering and Associated Lists.
MIL-C-3600	Compressors, Air and Gas (except Oxygen) Packaging of.
MIL-H-5440	Hydraulic Systems, Aircraft Types I and II, Design and Installation Requirements for.
MIL-R-5520	Reservoirs, Aircraft, Hydraulic, Non-separated Type.
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-C-5756	Cable and wire, Power, Electric, Portable (Asg).
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for.
MIL-A-8421	Air Transportability Requirements, General Specification for.
MIL-E-16298	Electric Machines having Rotating Parts and Associated Repair Parts, Packaging of.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-471	Maintainability Demonstration.
MIL-STD-808	Finish Materials and Processes for Corrosion Prevention and Control in Support Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-826	Electromagnetic, Interference Test Requirements and Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-2073/1A	DOD Materiel Procedures for Development and Application of Packaging Requirements.
MS25331	Light Assembly, Press to Test Indicator.
MS27265	Connector, Plug, Electrical and Ramp Power, 416/240 Volts, Male.
MS28720	Filter, Aircraft Hydraulic Line Type, 3,000 psi.
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure.
MS28764	Valve, Check, Controllable, Rotary-action, Hydraulic, 3,000 psi, Type 1i Systems (Asg).
MS28777	Washer, Flat, Aircraft Hydraulic Backup.
MS28778	Packing, Preformed, Straight Thread Tube Fitting Boss.
MS33649	Boss, Fluid Connection - Internalstraight Thread.
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal.

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MS33657	Fitting End, Standard Dimensions for Bulkhead Flared Tube Connections.
MS51335	Pintle Assembly, Towing - 18,000 lbs. Capacity, Manual Release.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

AIR FORCE - AERONAUTICAL STANDARDS

AN832	Union Flared Tube 3/8 Bulkhead and Universal.
AN910	Coupling, Pipe.
AN924	Nut, Tube, Bulkhead and Universal Fitting.
AN929	Cap Assembly, Tube, Pressure Seal.
AN6235	Filter Element - Hydraulic Replaceable Micronic Line Type.
AN6236	Filter Element and Hydraulic Replaceable Micronic Reservoir Type.
AN6249	Valve, 3000 PSI Hydraulic Check.
AN6270	Hose Assembly and Detachable Swivel Fitting, Low Pressure.
AN6289	Nut, Tube.

AIR FORCE -NAVY AERONAUTICAL DESIGN STANDARDS

AND10260	Flange-Type X Accessory Mounting.
AND10261	Flange-Type XI Accessory Mounting.
AND10266	Flange-Type XVI Accessory Mounting.
AND20006	Drive-Type XVI Engine Accessory.

DRAWING

AIR FORCE

48B7796	Ring Assembly - Tie Down 10000 Lb.
49B6564	Ring - Eyebolt 25000 Lb Cargo Tiedown.
55C6182	Connectors - 440V/416V Power Cable.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D-3951-82	Packaging, Commercial.

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(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MGI	Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlanta Ave, Boston, MA 02210.)

UNDERWRITERS LABORATORIES, INC (UL)

UL-674-82	Motors and Generators, Electric, for use in Hazardous Locations, Class I, Groups C and D, Class II, Groups E, F and G.

(Application for copies should be addressed to Underwriters Laboratories, Inc., 207 E. Ohio Street, Chicago, IL 60611.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Components. The stand shall consist of the following major components:

Item or system name	See requirement
Hydraulic system	3.9.1
Electrical system	3.9.2
Pneumatic system	3.9.3
Fan or blower	3.9.4
Trailer and housing	3.9.5

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3.3 General. The requirements of MIL-H-5440 for materials, fluids, fittings, reservoir venting, tubing, hose, pressure testing, and cleaning of parts and systems apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Materials.

3.5.1 Protective treatment. When materials are used in the construction of the stand that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.6 Design. The stand shall be designed to be a self-contained, mobile testing unit providing not less than 27 gpm flow at 2,750 psi with the components arranged as shown on figure 1. Components shall be arranged for maximum ease of operation and maintenance consistent with the versatility and efficiency required to conform to the requirements specified herein. The stand shall also be designed to provide 15 gpm at 4,600 psi.

3.6.1 Reliability. The stand shall have a minimum MTBF of 152 hours with a minimum reliability of 0.967 for a 5-hour mission at a confidence factor of 0.90.

3.6.2 Maintainability. Maintainability requirements shall be in accordance with MIL-STD-471.

3.6.3 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.7 Construction. The stand shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

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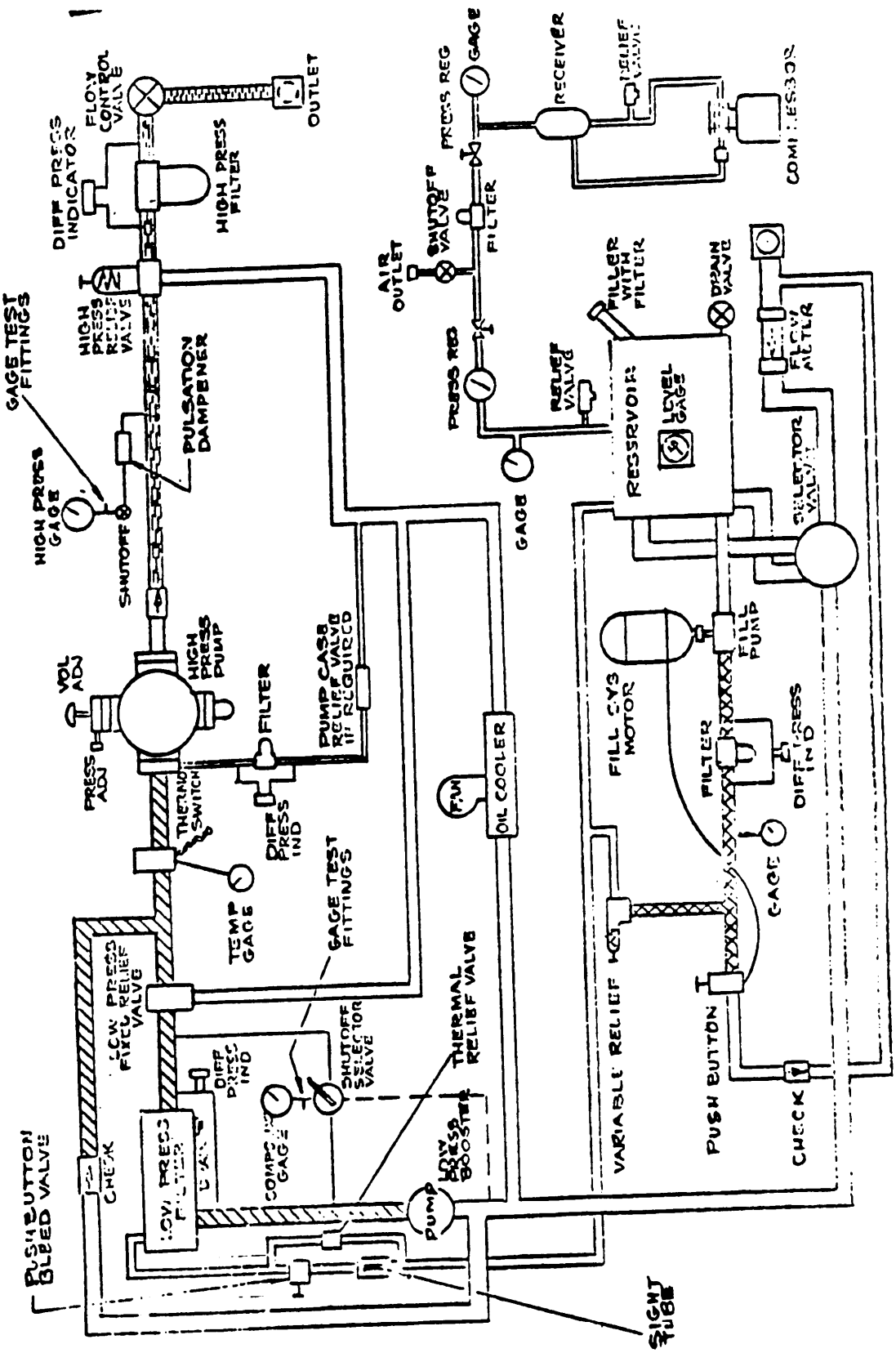


FIGURE 1. Schematic Hydraulic System Diagram.

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3.7.1 Stand. The stand shall be so constructed that adjustments, repair, and overhaul can be easily accomplished by the personnel of operating units and overhaul bases. The stand shall be so constructed that simple handtools may be used for maintenance and servicing and no special tools will be required. The stand shall be constructed to provide for maximum safety to personnel operating or maintaining the unit.

3.7.2 Forces and loadings. The stand structure, components, and mounting shall be constructed to withstand the gravitational forces and acceleration loadings specified in MIL-A-8421.

3.7.3 Operating controls. All operating and service controls shall be readily accessible to and operable by an operator wearing heavy gloves.

3.7.4 Fittings. All fittings and threaded bosses in the hydraulic circuits and components shall conform to MS33649, MS33656, or MS33657. All 5,000 psi pressure AN and MS fittings shall be steel. All bulkhead-type, high-pressure fittings shall use a locknut conforming to AN6289 with a backup gasket conforming to MS28777. All seals used on fittings shall conform to MS28778.

3.8 Performance. The stand shall be capable of operating under and withstanding the following conditions:

a. Operating temperatures ranging from -40° to $+ 125^{\circ}$ F.

b. Pressure altitudes from sea level to 6,000 feet.

c. A relative humidity of 95 to 100 percent at an operating temperature of $+ 125^{\circ}$ F and a storage temperature of $+ 160^{\circ}$ F.

d. Fungus growth as encountered in tropical climates.

e. Sand and dust particles as encountered in desert areas.

f. Rainfall as encountered in any locale.

g. Exposure to salt fog.

3.8.1 Storage temperatures. The stand shall be capable of withstanding storage temperatures ranging from -65° to $+ 160^{\circ}$ F.

3.8.2 Endurance. The stand shall be capable of withstanding 350 hours of operation between major overhauls.

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3.8.3 Operating positions. The stand shall be capable of functioning satisfactorily when its normal operating plane is at an angle of 8-1/2° in any direction from horizontal.

3.9 Details of components.

3.9.1 Hydraulic system.

3.9.1.1 High pressure pump. One axial-piston-type, high-pressure pump shall be provided. It shall have aircraft pump impulse characteristics by using nine pistons and high operational speeds (1,500 to 4,000 rpm inclusive). The pump shall be of the variable-stroke, variable-pressure, pressure-compensated type with adjustable maximum volume control. The pump shall be of sufficient rating to meet all requirements specified herein, but shall not be less than 30 gpm at 3,000 psig at 3,750 rpm and 15 gpm at 5,000 psig at 3,750 rpm.

3.9.1.1.1 Life expectancy. The high-pressure pump shall have a minimum life expectancy of 1,000 hours.

a. The pump shall require not more than 54 hp from the motor specified in 3.9.2.1 when being driven at 3,600 rpm and pumping hydraulic fluid conforming to MIL-H-5606 at 27 gpm and raising the fluid pressure from 40 to 3,000 psig at 68° F.

b. The pump shall be designed to withstand and operate throughout inlet fluid temperatures ranging from -40° to +165° F and inlet pressures of 40 to 300 psig.

3.9.1.1.2 Pressure compensator control. The pump shall have an adjustable pressure compensator control for regulating the pressure at a constant value with the flow varying from 0 to 27 gpm. The discharge pressure of the pump shall be 2,700 ± 50 psig when delivering 27 gpm and when the zero flow pressure setting is at 3,000 psig. The pump shall meet the performance characteristics shown on figures 2 and 3. The pump shall compensate from zero flow to full flow or vice versa in 0.05 second, or less, at any speed above 50 percent of rated speed and shall limit the maximum transient pressure during cutout to not more than 133 percent of nominal rated pressure. The pump pressure shall stabilize to a steady-state value within 10 seconds. The adjustment range of the pressure compensator shall be from 500 to 5,000 psi. The variable volume control and pressure control shall be mounted on or easily accessible from the control panel.

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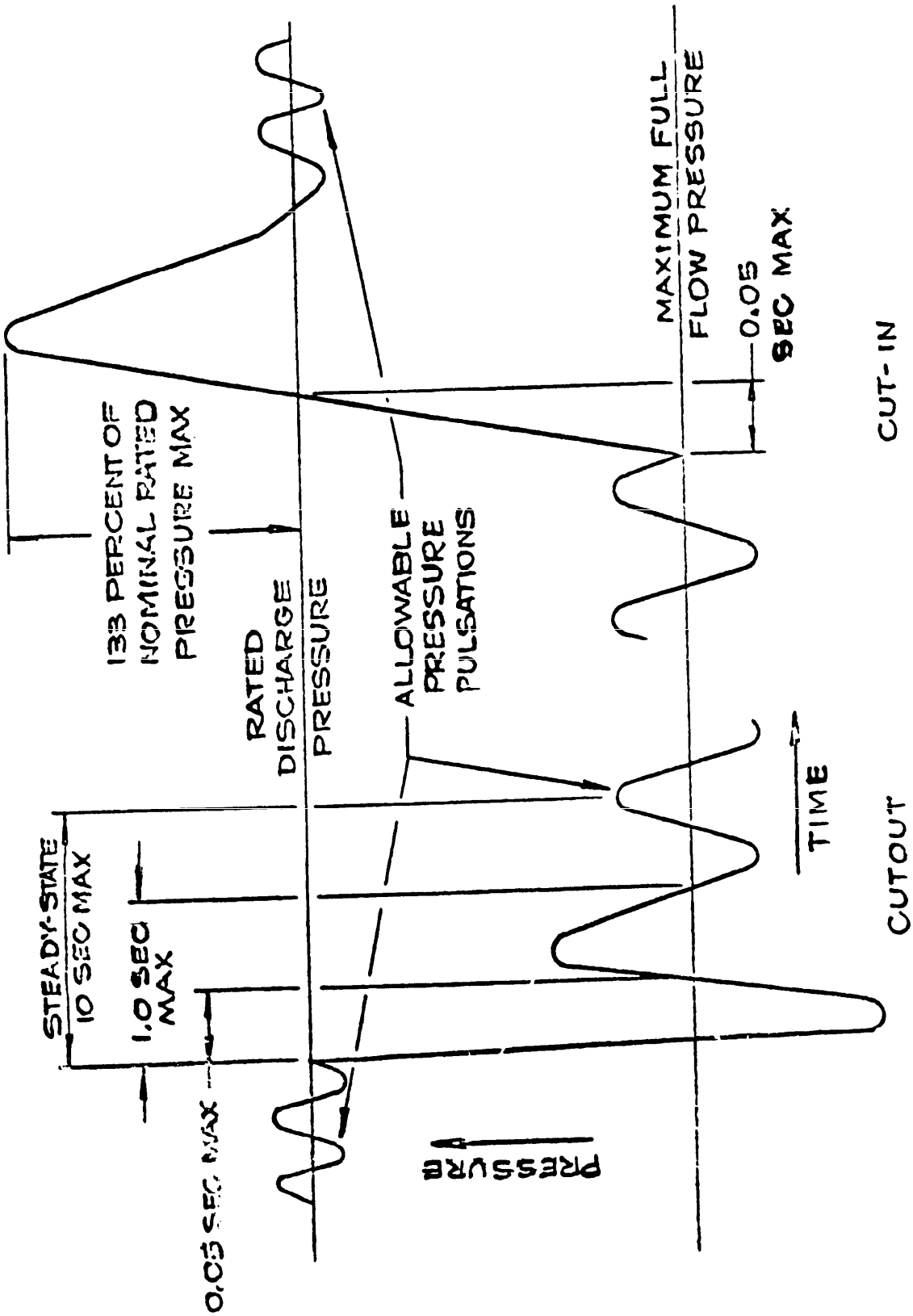


FIGURE 2. Typical pressure vs time chart.

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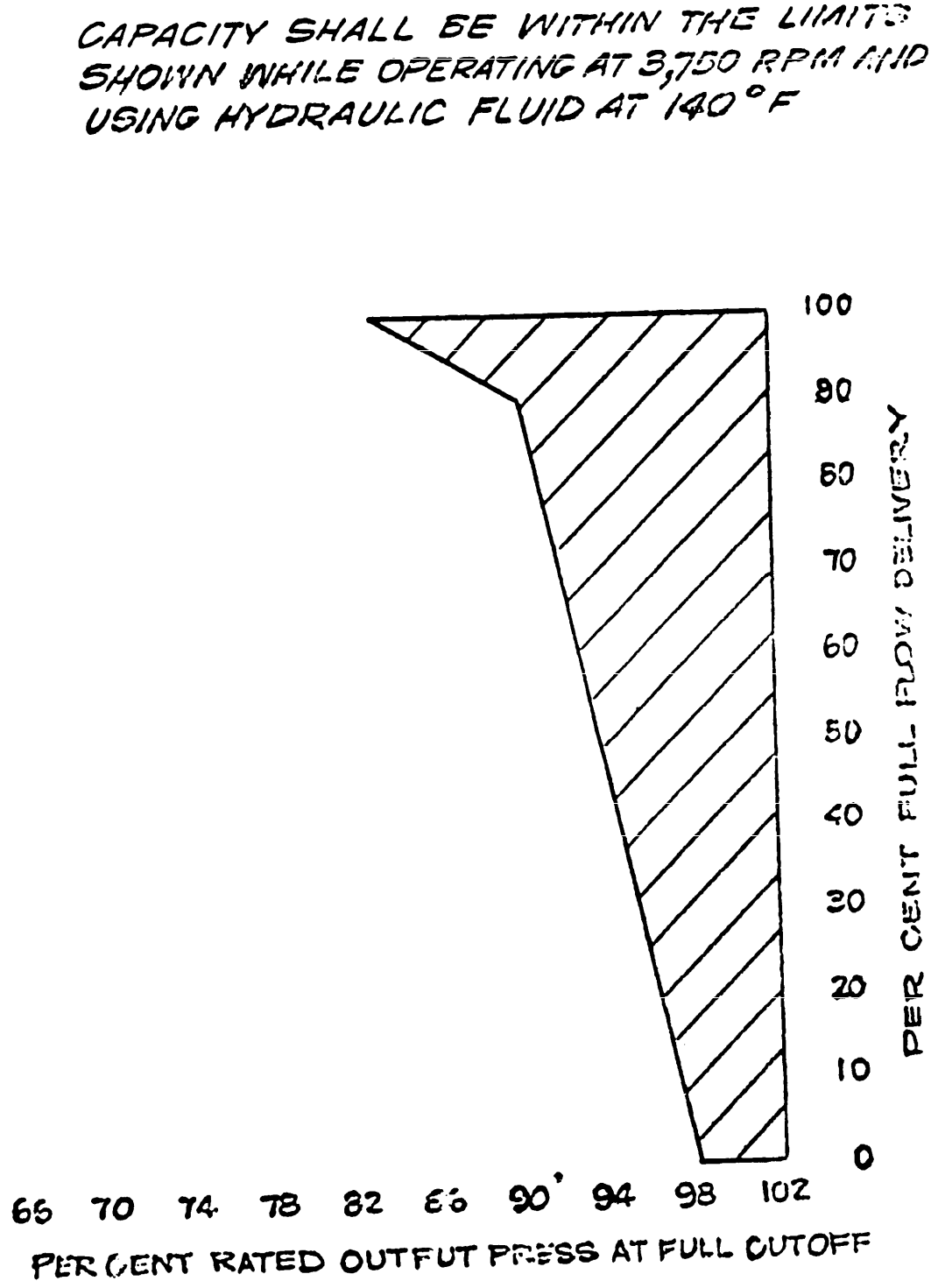


FIGURE 3. Flow characteristics.

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3.9.1.1.3 Relief valve. If required to maintain the pump case pressure at a value recommended by the pump manufacturer, a fixed relief valve shall be provided.

3.9.1.1.4 Flanges and spline drive. The pump shall be furnished with a flange and spline drive conforming to AND10266, type XVI-C. Rotation shall be counterclockwise as viewed from the shaft end.

3.9.1.1.5 High-pressure pump protection. A pressure differential switch shall be provided in the system. The switch shall actuate when the differential pressure between the inlet port and case drain line port of the pump drops below 30 psi, or as otherwise recommended by the high-pressure pump manufacturer. The switch actuation shall illuminate an MS25331-9 red indicator light on the control panel. The light shall be wired through a holding relay which holds the light on following a low differential pressure. The switch shall actuate only when the differential pressure drop occurs for 0.25 to 0.75 second. In addition to illuminating the indicator light for the faulty system, the switch shall automatically open the circuit to the main drive motor. The red light for the faulty system shall remain illuminated after the motor stops. Before restarting the faulty system, it shall be necessary to manually reset the device. The reset device shall be on the control panel. The check valve shall be located as shown on figure 1 and as specified in 3.9.1.13.

3.9.1.2 Low-pressure (boost) pump. One hydraulic boost pump shall be provided as shown on figure 1. The pump shall have a minimum life expectancy of 1,000 hours. The boost pump shall continuously supply at least 35 gpm to the high-pressure pump with the inlet pressure at 20 inches Hg vacuum and the outlet pressure not less than 40 psig nor more than 300 psig, as recommended by the high-pressure pump manufacturer, at all flow and pressure conditions specified herein. The 20 inches Hg vacuum with the above specified flows and pressures shall be required when the fluid temperature is below 20° F, and 15 inches Hg vacuum shall be required at all fluid temperatures above 20° F. The boost pump shall have sufficient output flow to provide additional fluid flow for filter bleed operations without substantially reducing the high-pressure pump inlet pressure. Check valves shall be located as shown on figure 1 to permit fluid to bypass the boost pump. The check valve shall be of the 1-1/2 inch, quick-acting, poppet type with a maximum of 2 psi pressure drop when passing 27 gpm.

3.9.1.2.1 Mounting flange. The boost pump shall be furnished with a flange and spline drive conforming to AND10261, type XI-C. Rotation shall be clockwise as viewed from the shaft end.

3.9.1.3 Filters. Air, fill systems, low pressure, high pressure, and pump case drain filters shall be provided and located in the systems as shown on figure 1. Each filter or set of filters shall be protected with a differential pressure switch installed as shown on figure 1 and shall indicate when the pressure drop across the filter reaches a value at which change of the filter element is required. The indication shall be the closing of a switch which illuminates an MS25331-9 red light on the control panel. The filters shall be located to permit easy draining of the case and replacement of the elements.

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3.9.1.3.1 High pressure filter. One high pressure filter with AN 6235 Filter Elements shall be incorporated as shown on figure 1.

3.9.1.3.2 Low pressure filter. A low pressure filter shall use two or more replaceable AN6236-3 elements. The low pressure filter shall be so located on a level above any other part of the line between the boost pump and high pressure pump that the air will collect within the filter and not be discharged into the high pressure pump inlet line.

3.9.1.3.2.1 Filter bleed valve. A pushbutton type bleed valve shall be mounted on the control panel and shall serve to bleed entrapped air to the reservoir from the top of the low pressure filter case or any other location at which air may be apt to collect. The valve shall also serve to bleed or drain the reservoir of the aircraft, when desired. Force necessary to operate this valve shall not exceed 10 pounds. When in the released position, the valve shall not permit leakage from the aircraft hydraulic system to exceed 2 drops in 5 minutes. Valve and tubing size shall be not less than 3/8 inch tube size.

3.9.1.3.2.1.1 Sight tube. A sight tube shall be furnished in the bleed line as shown on figure 1. It shall be mounted on the control panel adjacent to the bleed valve. The sight tube shall withstand a 200 psi working pressure and a 300 psi proof pressure. It shall be 3/8 inch size and 3 inches minimum visible length. The sight tube shall be lighted from behind for easy visibility of contamination or entrained air.

3.9.1.3.3 Pump case filter. A filter conforming to or dimensionally interchangeable with an MS28720-12 filter, suitable for 125 psi pressure, containing AN6235-4A elements shall be provided in the pump case drain line as shown on figure 1.

3.9.1.4 Thermal pressure relief valve. A preset relief valve dimensionally interchangeable with check valves conforming to AN6249 shall be provided in the hydraulic system as shown on figure 1. The valve shall relieve pressures in excess of 275 psi and thus preclude damage to components in the low pressure hydraulic circuit.

3.9.1.5 Relief valves. High pressure and low pressure relief valves shall be provided and located in the system as shown on figure 1. The valves shall be located as near the high pressure pump inlet and outlet as possible. The relief valves shall not be located on the control panel, but shall be easily accessible. The relief valves shall incorporate a locking device so that vibrations cannot change their settings.

3.9.1.5.1 High-pressure relief valve. The high-pressure relief valve shall be adjustable by hand from 500 to 5,500 psig.

3.9.1.5.2 Low-pressure relief valve. The low-pressure relief valve shall be adjustable to a fixed pressure of not less than 50 nor more than 150 psig. The low-pressure relief valve shall close to zero flow in 0.025 second and shall have a pressure drop no greater than 25 psi when passing 27 gpm.

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3.9.1.6 Reservoir. A stainless steel or aluminum hydraulic fluid reservoir shall be mounted within the stand enclosure. The reservoir shall have a sufficient capacity to supply at least 15 gallons of fluid to the system when inclined 8-1/2° from vertical in any direction. The reservoir shall have additional space for a 3 gallon expansion above the normal full level and at least a 2 gallon space below the outlet to the system for collecting moisture and impurities. The reservoir shall be designed for operating pressures of 50 psi and structural proof tests of 100 psi. The reservoir shall conform to MIL-R-5520 regarding obstruction to flow, structural design, shell, seams, structural strength, baffles, fluid connections, sump, drain, return line inlet, scuppers, filter unit, finish, and filling time. Sufficient clearance shall be provided around the filler opening to permit rapid filling from 1-gallon cans while the stand is in operation. The reservoir fill cap shall be either the type that cannot be removed while the reservoir is pressurized, or the type that will automatically depressurize while it is being removed.

3.9.1.6.1 Access plates. One or more access plates of at least 8 x 6 inches in accordance with standard design practices for ASME coded vessels shall be provided to permit visual inspection and manual cleaning of the interior of the reservoir.

3.9.1.6.2 Drain valve. A drain valve shall be provided for completely draining the reservoir. Necessary clearance shall be provided around the drain outlet to permit rapid draining into 1-gallon containers.

3.9.1.6.3 Fluid level indicator. A dial type fluid level indicator shall be provided and shall indicate the quantity of available fluid in the reservoir to an accuracy of within ± 2 gallons of actual quantity. The indicator shall be mounted on the instrument panel.

3.9.1.6.4 Reservoir selector valve. A 4 way, 2 position selector valve shall be provided (see figure 1) to either route the return fluid to and from the stand reservoir or to cause the return fluid to bypass the reservoir. The valve shall be designed to operate in pressures varying between 20 inches Hg vacuum and 200 psi. The valve shall be mounted on the instrument panel. Internal leakage shall not exceed 5 drops per minute under any operating condition.

3.9.1.7 Stand outlet connections. Outlets shall be provided as shown on figure 1. The connections shall not be located where leakage of fluid from the connection could fall on the trailer tires or within the enclosure of the stand.

3.9.1.7.1 Pressure connections. The pressure outlet and dust cap shall be equal to and dimensionally interchangeable with Aeroquip part No. 015628-S2-16 and 155-S7-16D, respectively. All valves, connections, and tubing used from the high-pressure pump discharge to the end of the 10-foot pressure hose shall be of a size that will not permit the pressure drop to exceed 60 psi with 27 gpm flow at 120° F fluid temperature. All connections and fittings on the high-pressure side shall be not less than 1-inch tubing size. Tubing shall be used on all high-pressure lines within the stand, except as otherwise specified or where impractical because of vibration. The entire high-pressure portion of the systems shall be designed for proof pressures of 7,500 psig.

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3.9.1.8 Return connection. A return connection shall be provided as shown on figure 1. The connection shall conform to Aeroquip Part No. 145-S-24D or be equal to and dimensionally interchangeable with the Aeroquip part. A dust cap conforming to the Aeroquip Part No. 145-S2-24D or equal to and dimensionally interchangeable with aeroquip part shall be attached to the outlet (see 3.9.1.17.1). The size of all the connections and components used in the return side of the stand hydraulic system shall be such as to permit not more than a 10-psi drop at 27 gpm with fluid temperature of 120° F from the aircraft outlets through 10 feet of hose to the suction side of the boost pumps. Return lines and components and low pressure lines and components shall be not less than 1-1/2 inch tubing size.

3.9.1.8.1 External and internal leakage. Total external leakage shall not exceed 2 drops per hour at any flow or pressure specified herein. Internal leakage shall not exceed 5 cubic inches per hour from the closed loop system.

3.9.1.9 Oil cooler. An air-to-oil oil cooler shall be provided in the system as shown on figure 1. The cooler shall be capable of maintaining the oil temperature of the system below +160° F when 27 gpm at 750 psig is by-passed through the cooler with an ambient air temperature of +125° F. The cooler shall permit full flow through the bypass without damage resulting therefrom. The oil cooler shall be capable of operating under a pressure of 200 psi without leakage or damage. Means shall be provided to make the cooler inoperable in low temperature. Provisions to accomplish this shall be accessible but need not be on the control panel.

3.9.1.10 Flow control valve. A 1-inch valve or larger shall control or stop the flow into or from the pressure outlet. The valve shall be rated for 27 gpm flow and 5,000 psi pressure. The valve shall accurately meter the flow throughout the volume range of 0 to 27 gpm and shall be so pressure-balanced that a force of 40-inch pounds is the maximum required turning force at 3,000 psi. The valve shall be mounted on the control panel and shall be capable of going from the closed position to the open position in two full turns of the handle. The valve shall be capable of controlling the flow within ±1/4 gpm when flow rates are below 5 gpm, and ±1 gpm with flow rates of from 5 to 27 gpm.

3.9.1.11 Bypass valve. A bypass valve, identical to the flow control valve required in 3.9.1.10, shall be provided in the system as shown on figure 1. The valve shall be mounted on the control panel.

3.9.1.12 Fill system. A system incorporating an electric-motor-driven pump, filter, puchbutton-type fill valves, pressure gage, and relief valves as shown on figure 1 shall be provided to add oil to the aircraft or stand system as desired.

3.9.1.12.1 Filling pump and motor. The filling pump and motor combination shall pump hydraulic fluid at a minimum rate of 2 gpm at 150 psig with fluid temperature of 20° F. The pump motor shall be a 200/440-V, 3-phase, 60-cycle motor of sufficient rating to continuously operate the pump at maximum rated pressure and output without exceeding safe temperature limits for the class of insulation used. Overload relays shall be provided for protecting the pump motor. The pump motor shall be connected as shown on figure 4. The pump shall be furnished with flange and spline drive conforming to AND10260, type X. The pump shall be driven at 3,600 rpm.

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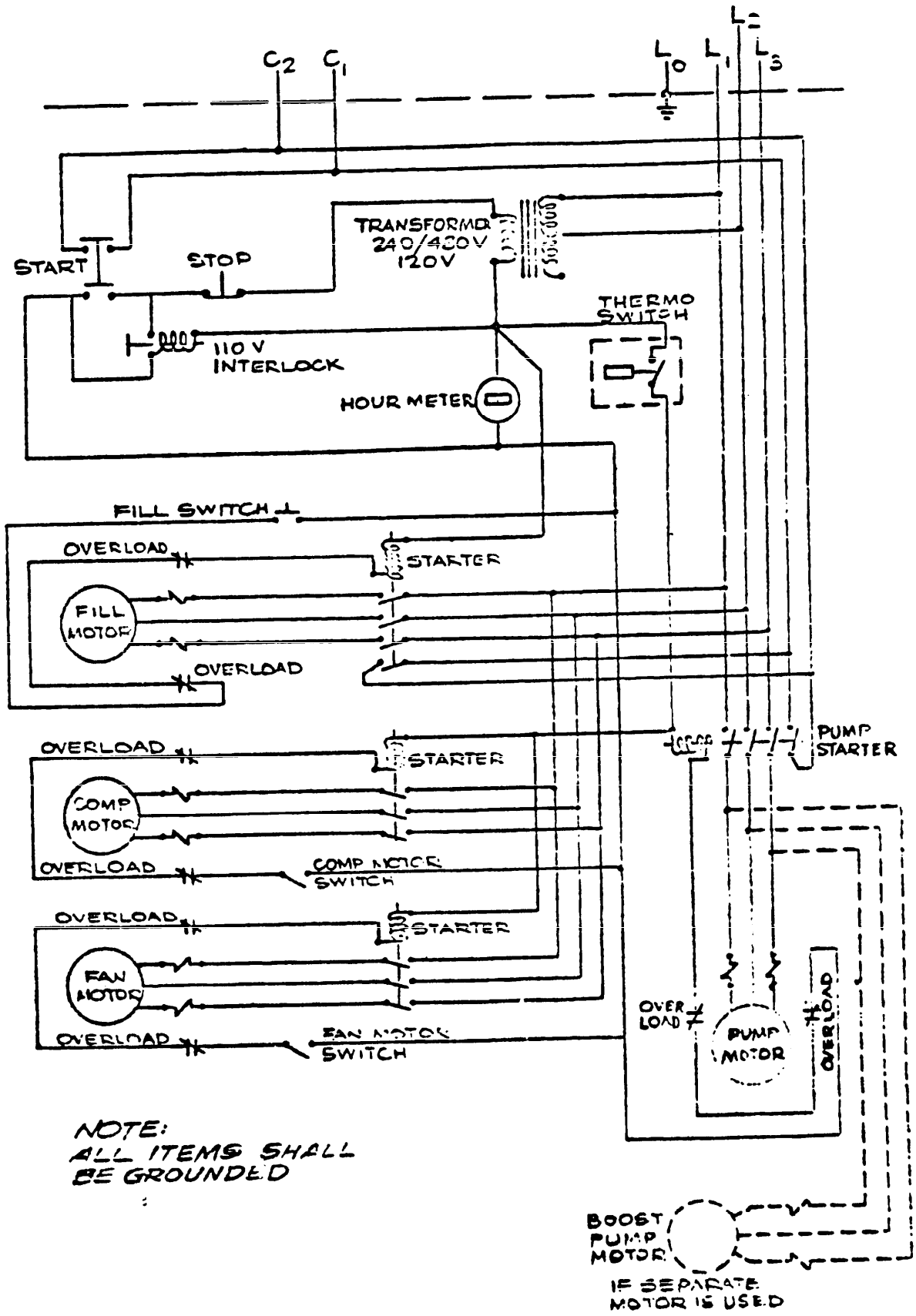


FIGURE 4. Electrical system diagram.

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3.9.1.12.2 Filling valve. A pushbutton type filling valve shall be mounted on the control panel. Force necessary to operate the valve shall not exceed 10 pounds. Actuation of the valve shall simultaneously open the hydraulic circuit and start the fill motor. When in the release position, the valve shall not permit leakage from the aircraft hydraulic system to exceed 10 drops in 5 minutes. Valve and tubing size shall be not less than 3/8 inch tube size.

3.9.1.12.3 Fill system relief valve. One variable relief valve as shown on figure 1 shall be mounted on the control panel. The valve shall be adjustable throughout a pressure range of from 0 to 140 psi and capable of relieving the total volume discharged by the fill pump. The valve shall have a locking device so that vibrations cannot change the pressure setting. The relief valve shall be so designed that maximum pressure setting will be 140 $\pm$ 5 psi.

3.9.1.12.4 Fill system check valves. Two check valves dimensionally interchangeable with MS28764 valves, or larger, shall be located in the fill system as shown on figure 1 to prevent reverse flow to the fill system pump.

3.9.1.12.5 Fill system filter. A filter shall be provided in the fill system as shown on figure 1. The filter shall be an AN6235-4A filter element in a filter conforming to or dimensionally interchangeable with an MS28720-6 filter suitable for 145 psi operating pressure.

3.9.1.12.6 Fill system connections. All lines, tubing, hoses, and connections used in the fill system shall be 3/8 inch tubing size, or larger.

3.9.1.13 Pressure check valve. A pressure check valve dimensionally interchangeable with the AN6249-16 valves, but suitable for operation at 5,000 psi, shall be located as near the high pressure pump outlet as practical (see figure 1) to prevent reverse flow to the high pressure pumps. Reverse flow shall not exceed 2 drops per minute at any pressure from 100 to 5,000 psi.

3.9.1.14 Flowmeter. A variable-orifice-type visual flowmeter rated at 300 psig shall be provided in the return line as shown on figure 1. The flowmeter shall be accurate within $\pm$ 2 percent of full scale reading. The flowmeter shall be lighted from behind so fluid contamination or air is clearly visible.

3.9.1.15 Gages.

3.9.1.15.1 Hydraulic gages. All hydraulic gages shall conform to GG-G-76, except where conflicts exist this specification shall govern. All hydraulic gages shall be of the recess type with a mounting flange at the front of the gage. The mounting flange and dial shall not extent more than 5/8 inch beyond the external surface of the panel. The gages shall be mounted by screws and tapped panel holes shall be connected with flexible AN or MS hose, or commercial equivalent, to permit removal and installation by one man from in front of the panel. The high pressure gage line shall be provided with a needle shutoff valve and gage test connection to permit calibration of the gage in the panel. Means shall be provided to restore the accuracy of the high-pressure and low-pressure gages without removal from the panel. All gages shall be designed for operation at -40° to $+ 180^{\circ}$ F. The hydraulic gages shall

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be corrosion-resistant and shall incorporate good quality stainless-steel movements and steel alloy bourdon tubes. The dial and pointer shall be separated from the gage movement by a wall which will prevent rupture through the front of gage failure. The gage cases shall incorporate a safety blowout device that will automatically relieve the case pressure should it reach 1/2 psi.

3.9.1.15.1.1 High-pressure gage. A high-pressure gage with a dial diameter of 6 inches and a dial range of 0 to 6,000 psi shall be provided. The gage shall be accurate to within ±1/2 percent, or less, of full scale at all operating temperatures specified herein. The high-pressure gage shall be protected by an accumulator-type hydraulic dampener.

3.9.1.15.1.2 Low-pressure gage. A gage having a dial diameter of 4-1/2 inches, or larger, and a dial range of 30 inches Hg vacuum to 300 psi pressure shall be provided. The gage shall be accurate to within 1 percent of full-scale range.

3.9.1.15.2 Shutoff valve. A needle shutoff valve for the high-pressure gage shall be of the panel-mounted type with corrosion-resistant steel stem. The valve shall be capable of withstanding pressures up to 5,000 psi.

3.9.1.15.3 Gage test fittings. Gage test fittings shall be provided for the high and low pressure gages. The following fittings shall be mounted on the control panel adjacent to their respective gages:

a. AN832-4D nipple.

b. AN924-4D nut.

c. AN929-4 cap.

3.9.1.15.4 Gage selector valve. A shutoff-type selector valve shall be mounted on the instrument panel and shall permit pressure selection at the various points as shown on figure 1. One position of the selector shall isolate the low-pressure gage for calibration.

3.9.1.15.5 Temperature gage. A flush-mounted, remote-reading temperature gage shall be provided. The range of the gage shall be approximately -20° to +200° F and shall indicate within ±5° F of the temperature of the fluid supply to the high-pressure pump. The gage shall have a 3-inch diameter dial, or larger. The dial face of the temperature gage shall be marked with a red sector as a warning at 160° F and higher. The accuracy of the gage shall not be affected by continued exposure to the climatic extremes specified herein.

3.9.1.15.6 Fill pressure gage. A corrosion-resistant gage with a 2-1/2 inch dial face shall be mounted on the stand panel to indicate the pressure in the fill system. The gage shall have a range of from 0 to 160 psi.

3.9.1.16 Thermoswitch. An adjustable-type preset thermoswitch shall sound a warning horn or bell when the oil inlet temperature to the high-pressure pump exceeds +165° F. The horn or bell shall be audible for a distance of 25 feet from the stand when the stand is operating at full capacity. The thermoswitch shall have an adjustable range of from +150° to +200° F.

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3.9.1.17 Hose assemblies. The hose assemblies specified in Table I shall be provided with each stand and shall be equipped as specified.

3.9.1.17.1 Dust plugs and caps. All dust plugs and caps shall be attached to the stand and hose assemblies by metal chains.

3.9.1.17.2 Adapters. The following adapters shall be furnished in the stowage compartment.

3.9.1.17.3 Couplings.

a. Two each coupling halves, Aeroquip Part No. 015630-S2-16 and No. 155-S5-123D or equal to and interchangeable with the Aeroquip parts, shall be fastened into two double-threaded couplings similar to those conforming to MS 33649-16 threads in one end and MS33649-12 threads in the other. The threaded coupling shall be not more than 3 inches long and shall be capable of withstanding pressures of 5,000 psi and proof pressures of 7,500 psi.

b. Two each coupling halves, Aeroquip Part No. 014519-S2-24D and No. 145-S5-16D or equal to and interchangeable with the Aeroquip parts, shall be fastened into two double-threaded couplings similar to those conforming to AN910 with MS33649-24 threads in the other. The threaded couplings shall be not more than 3 inches long and shall be suitable for working pressures of 200 psi and proof pressures of 300 psi.

3.9.2 Electrical system. All electrical equipment and wiring shall conform to NFPA 70. All electric motors shall conform to UL-674-82. All electrical equipment less than 18 inches from the ground shall be explosion-proof for class I, division 2 location in compliance with article 500 of NFPA 70. The electrical system shall be designed for operation on a 220-V, 3-phase, 60-cycle power supply and shall be designed for conversion to 440-V, 3- phase, 60-cycle power without replacing any major component. The voltage tolerance shall be ±10 percent and the frequency ±5 percent. The electrical system shall be connected as shown on figure 4.

3.9.2.1 Pump motor. A design B or C, code D or E, ball-bearing, squirrel-cage induction motor of not less than 60 hp continuous rating conforming to NEMA MG 1 shall be provided for driving the high and low-pressure pumps. The pump-motor shall conform to UL-674-82 and shall be of sufficient rating to operate the stand continuously under the conditions specified herein at maximum capacity without exceeding the NEMA recommended temperature limits for the class of insulating materials used. The manufacturer's recommended full -load current shall not be exceeded during operation. The motor shall operate at a pressure altitude of 6,000 feet. Means shall be provided to reduce starting current surges to 100 amps when connected for 440-V operation and to 250 amps when connected for 220-V operation. The high-pressure pump shall be mounted on the motor on mountings in accordance with AND20006.

3.9.2.2 Pump motor control. The stand shall be provided with a remote-operated magnetic motor control system as shown on figure 4, providing motor overload protection and circuit protection. The remote-operated control shall be powered by 110V, obtained by reducing input voltage with a transformer. Voltages greater than 120V shall not be used on the control panel.

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TABLE I

Hose Assemblies

Hose Assembly Part No.	Quantity	Hose Ends (One or Both)	Equipped With
Precleaned Teflon, lined 1 inch comm'l hose, 3,000 psi working 6,000 psi proof pressure (10-foot length)	1 ea	Both	Coupling half, Aeroquip part No. 015828-S5-16* Dust plug, Aeroquip part No. 155-S9-16D*
MS28741-24-1200 (10-foot length)	1 ea	Both	Coupling half, Aeroquip part No. 145-S5-24D* Dust plug, Aeroquip part No. 145-S9-24D*
Precleaned Teflon lined 1 inch comm'l hose, 3,000 psi working 6,000 psi proof pressure (10-foot length)	1 ea	One	Coupling half, Aeroquip part No. 015828-S5-16* Dust plug, Aeroquip part No. 155-S9-16D*
		Other	Coupling half, Aeroquip part No. 015630-S2-16* Dust cap, Aeroquip part No. 155-S7-16D*
MS28741-24-1200 (10-foot length)	2 ea	One	Coupling half, Aeroquip part No. 145-S5-24D* Dust plug, Aeroquip part No. 145-S9-24D*
		Other	Coupling half, Aeroquip part No. 014519-S2-24D* Dust cap, Aeroquip part No. 145-S7-24D*

* Or a part equal to and interchangeable with the Aeroquip part.

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3.9.2.2.1 Starters and starter case. A starter of sufficient capacity to withstand openings under starting surges shall be provided. A starter case of sufficient size to enclose all remote controls (contractors, overload devices, transformers, starters) for all electric motors shall be furnished. The starter shall include two overload relays for motor protection which can be manually reset without opening the starter case. A 240/480-V to 120-V control transformer shall be furnished in the starter enclosure.

3.9.2.2.2 Pushbutton. A start-stop, momentary contact, low voltage protection pushbotton shall be mounted on the instrument panel. The start button shall incorporate a cutout for the boost pump interlock system specified in 3.9.1.1.5.

3.9.2.2.3 Circuit breaker. A circuit breaker rated in accordance with NFPA 70 shall be provided for each circuit or circuit branch.

3.9.2.2.4 Wiring. All wiring between electrical components shall be in accordance with NFPA 70. A color code or number code shall be used on all wires and terminal lugs to simplify tracing and reworking of the circuits.

3.9.2.2.5 Hourmeter. A counter-type hourmeter having a range of from 0 to 9,999 hours shall be provided to record the hours of operation of the pump motor.

3.9.2.2.6 Electric cable and connector. The alternatinf-current input cable shall be an extra-flexible, 6 conductor cable in accordance with MIL-C-5756, 50 feet in length. The cable shall use three conductors not smaller than AWG No. 00 connected to contacts L_1, L_2, L_3, one conductor not smaller than No. 8 connected to contact L_0 (ground) and two conductors not smaller than No. 12 connected to C_1 and C_2. The connector shall be a water-tight, molded rubber, polarized type conforming to MS27265. The cable shall be securely attached to the stand to remove all strain from the terminals.

3.9.2.2.7 Reverse rotation protection. A mechanical or electrical device shall be provided for preventing operation of the pump motor in the opposite direction of rotation from that required for the pumps, in the event of incorrect phasing,

3.9.3 Pneumatic system. A compressed air system shall be provided for pressurizing aircraft reservoirs up to 150 psig or the unit reservoir up to 50 psig.

3.9.3.1 Air compressor. An air compressor delivering 3 cfm at 150 psi continuously when supplied with air at 77° F and 29.92 inches Hg absolute shall be provided.

3.9.3.2 Unloader valve. A pilot-actuated unloader valve shall sense pressure at the air receiver and shall unload the compressor when the receiver is fully charged.

3.9.3.3 Pressure regulators. Two pressure regulators designed to operate at -40° to +125° F ambient temperatures shall be mounted on the control panel. The pressure regulators shall control and limit the pressure to the outlet at any setting of from 5 to 150 psig and the unit reservoir pressure from 5 to 50 psig and shall maintain these set pressures at any flow condition including zero flow.

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3.9.3.4 Air pressure gages. Two 2-inch minimum diameter pressure gages, having a scale range of 0 to 200 psi, shall be included to indicate pressure in the aircraft reservoir and stand reservoir. The gages shall be accurate to with ±5 percent of full-scale range.

3.9.3.5 Relief valves. Fixed relief valves set at 160 and 60 psig shall be installed in the system to protect the outlet and reservoir pressurization systems.

3.9.3.6 Reservoir pressurization valve. A 3-way, 2-position valve shall be installed between the pressure regulator and stand reservoir. In one position the reservoir shall be pressurized and in the other position the reservoir shall be exhausted and the pressure regulating outlet blocked off.

3.9.3.7 Air receiver. An air receiver with a minimum capacity of 0.25 cubic feet shall be provided. The air receiver shall have a minimum working pressure of 150 psig and proof pressure of 300 psig. The air receiver shall be provided with a drain and sufficient clearance to drain collected water to the exterior of the stand.

3.9.3.8 Air filter. A filter conforming to or dimensionally interchangeable with an MS28720-12 filter, suitable for 150 psi pressure, shall be provided in the air system. The filter shall contain an AN6235-4A element.

3.9.3.9 Air Outlet. An AN832-4D nipple, AN924-D nut, and AN929-4 cap shall be provided for the outlet connection. The outlet connection shall be located adjacent to the hydraulic outlets. A shutoff valve shall be provided on the control panel to shutoff air to the outlet connection. A 30-foot length of AN6270-4-360 hose shall be provided.

3.9.3.10 Compressor motor. A 220/440-V, 3-phase, 60-cycle motor shall be provided. It shall be of sufficient rating to operate the compressor at maximum rated pressure and output continuously without exceeding safe temperature limits for the class of insulation used. The compressor motor shall be connected as shown on figure 2.

3.9.3.11 Compressor motor controller. A manual or magnetic push-button, 220/440-V, 3-phase, 60 cycle controller with two overload relays shall be provided for starting, stopping, and protecting the compressor motor. The compressor control shall be mounted on the control panel in an instrument cluster of all pneumatic controls and gages.

3.9.3.12 Hydraulic motor and control valve. A system, including hydraulic motor and control valve, may be used in place of the electric motor and control specified in 3.9.3.10 and 3.9.3.11.

3.9.4 Fan or blower. A fan or blower with a drive mechanism from the pump motor or driven by a separate electric or hydraulic motor shall be furnished. The fan shall provide sufficient cooling air to the air-to-oil cooler to meet all oil cooling requirements. The fan or blower shall normally function whenever the pumps are operating, however, for low temperature operation, provision shall be made to disconnect the drive mechanism for the fan or blower. The control switch shall not be mounted on the control panel, but instructions as to its location and operation shall be provided as part of directions for low temperature operation as specified in 3.17b. Should an electric motor drive be used, the requirements of 3.9.2, 3.9.3.10, and 3.9.3.11 shall govern motor and control design.

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3.9.5 Trailer and housing. The components of the stand shall be assembled and mounted on a trailer and covered with a weather-resistant housing. The stand shall meet the requirements of MIL-M-8090 for type II, group C mobility and shall meet the air transportability requirements of MIL-A-8421 in general class aircraft.

3.9.5.1 Trailer. The trailer shall be of the 4 wheel, 2 wheel-steerable type.

3.9.5.1.1 Frame. The trailer frame shall be of welded-steel construction. The frame shall be not less than 42 inches nor more than 54 inches wide and shall be not less than 60 inches nor more than 90 inches long. The ratio of width divided by length shall not exceed three-fourths.

3.9.5.1.2 Springs. The springs on the trailer shall provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be of such construction that not more than 3 inches is added to the frame height for spring deflection.

3.9.5.1.3 Parking brakes. Parking brakes in accordance with MIL-M-8090 shall be provided. Brakes for both rear wheels shall be interconnected and shall operate from one handle located near the tow bar.

3.9.5.1.4 Reflectors. Reflectors required by MIL-M-8090 need not be provided.

3.9.5.1.5 Lifting and tiedown fittings. Lifting and tiedown fittings conforming to Drawing 48B7796 or 49B6564 shall be provided and shall withstand the loadings specified in MIL-A-8421. The unit shall be designed for fork-lifting.

3.9.5.1.6 Pintle hook. A pintle hook conforming to MS51335 shall be attached to the back of the trailer.

3.9.5.2 Control panel. A control panel shall be provided for mounting the meters, valves, indicators, gages, switches, and controls furnished with the stand. The panel shall be constructed of No. 12 U.S. gage (0.1046 inch) sheet steel, or thicker, and shall be supported by a welded-steel, self-supporting framework.

3.9.5.2.1 Shock mounting. Instruments subject to error from vibration shall be shock mounted.

3.9.5.2.2 Grouping and color-coding of controls and gages. In each system, the fill button, fill system relief valve, bleed button, and sight tube shall be grouped together. The pneumatic system controls and gages shall be grouped together on one section of the control panel. The filter warning lights shall be grouped together in a conspicuous section of the panel.

3.9.5.2.3 Location and position. The control panel shall be located at a sufficient height above the ground and inclined at a sufficient angle (10° minimum) to facilitate convenient observation and operation by an operator approximately 5 feet, 10 inches tall. The control panel frame shall be bolted or welded to the trailer bed and so positioned that the trailer wheels, tow bars, et cetera, will not interfere with the operator.

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3.9.5.3 Housing. The stand components and control panel shall be completely enclosed and protected by a weather-resistant housing. The housing shall consist of a welded-steel framework covered with No. 16 U.S. gage (0.0598 inch) sheet steel, or heavier, with doors and adjustable louvers as required. The housing shall be adequately braced to prevent distortion or excessive vibration during handling and operation of the stand or handling of the housing while removed from the stand. The stand shall be so designed that water or hydraulic fluid spilled within the housing will drain to the exterior and will not collect within the enclosure.

3.9.5.3.1 Hooks. Metal hooks shall be provided on the exterior of the housing to permit temporary storage of the hoses. These hooks shall be designed to permit flat positioning against the housing when not in use.

3.9.5.3.2 Internal accessibility and ventilation. The housing shall be provided with sufficient access and ventilating doors and adjustable louvers to permit access for operating and servicing and to ventilate the stand. The doors and louvers shall be free from binding and shall be easy to operate with one hand. Provisions shall be incorporated to rigidly secure the doors in the full-opened or full-closed position.

3.9.5.3.3 Housing removability. The housing shall be easily removable by actuating not more than 10 quick-acting disconnecting fasteners and lifting or sliding the entire housing off the remainder of the stand assembly. Handles shall be provided for this purpose.

3.9.5.3.4 Housing supportability. The housing shall be capable of supporting a 200 pound weight evenly distributed over an area 4.5 by 5 inches and applied in any location on the top of the stand.

3.9.5.4 Compartments. A handbook compartment, 2 by 9 by 12 inches or larger, and an enclosed accessory box, 8 by 8 by 24 inches or larger, shall be permanently and securely attached at an accessible point within the housing. Provisions shall be made for storage of the hose assemblies and accessories specified herein within the outline of the housing. The compartments may be constructed on the trailer bed or may be fastened to the housing assembly.

3.10 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.11 Noise level. The noise level of the stand shall not exceed the values in the frequency ranges specified in 4.6.14.

3.12 Radio interference. The stand shall meet the requirements of MIL-STD-826, class 1B from 0.15 through 150 mc (radiated).

3.13 Dimensions. The length and width of the stand and housing shall be not less than the trailer frame assembly nor extend beyond the trailer wheels at any point. The overall height of the stand shall be held at a minimum consistent with good design practices, but shall not exceed 40 inches above the top of the tires.

3.14 Weight. The dry weight of the stand, including the trailer assembly, shall not exceed 4,000 pounds (without hydraulic fluid, et cetera).

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3.15 Color. Color shall be in accordance with MIL-STD-808, exterior code DG, interior code LG.

3.16 Finishes and protective coatings.

3.16.1 Cleaning, painting, and plating. Cleaning, painting, and plating shall be in accordance with MIL-STD-808.

3.16.1.1 Corrosion - resistant coatings. Tubing and components that have been treated with corrosion-resistant coatings need not be painted.

3.17 Operational markings.

a. When required, a caution plate containing the following information shall be permanently mounted on the instrument panel adjacent to the selector valves:

DO NOT OPERATE WHEN THE FLOW CONTROL VALVE IS OPEN.

b. A permanently engraved plate containing low temperature adjustment directions for the oil coolers shall be provided on the control panel.

c. An engraved or etched, permanent-type correction chart shall be provided near the flowmeter.

d. A permanent-type nameplate containing complete instructions for converting the stand from operation on 220V to 440V, or vice versa, shall be fastened inside the starter case specified in 3.9.2.2.1.

e. The needle shutoff valves shall have the manufacturer's cold-working pressure rating stamped on the valve body with an arrow indicating the flow direction.

f. The trailer parking brake control device shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be 1/4 inch black letters and shall be visible to the operator.

g. All items mounted on the control panel shall be identified as to use and cautions by permanently etched or engraved metal or plastic plates attached to the panel. Directions for increasing and decreasing flow, et cetera, shall be indicated for all controls.

h. An 8 by 10-inch, engraved, permanent-type diagram plate arranged similar to figure 1 to show the hydraulic circuit of the stand shall be securely fastened to the center of the instrument panel.

i. The following information shall be stenciled in 1-inch black letters on each of the four sides of the stand:

FOR AIR SHIPMENT THIS STAND IS DESIGNED FOR 8G FORE AND AFT, 3 G SIDEWARD.

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j. The following information shall be marked in 1-inch black letters on the drawbar or front of the enclosures:

MAX SPEED 20 MPH

k. The exterior of the stand shall be stenciled to indicate the pressure, center-of-gravity, and other markings necessary for proper handling and maintenance.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following additional marking shall be included:

RATING: 0 TO 27 GPM TO 3,000 PSI EACH SYSTEM, 0 TO 54 TOTAL.
15 GPM TO 5,000 PSI EACH SYSTEM, 0 TO 30 TOTAL.

TOWING SPEED: 20 MPH MAX.

3.19 Workmanship. The stand, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, bolts; et cetera.

3.19.1 Riveting. Riveting operations shall be performed to insure tight and satisfactory heading of rivets.

3.19.2 Cleaning. The stand shall be thoroughly cleaned. Loose, spattered, or excess solder, metal chips, and other foreign materials shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the stand shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance test (see 4.5).

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4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Unless otherwise specified, tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 70° F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Test period. Unless otherwise specified herein, any test period shorter than 1 hour shall not be credited as test time.

4.3.3 Apparatus and instruments. Apparatus and instruments used in conjunction with the testing specified herein shall be of laboratory precision-type and shall be checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation and shall be calibrated at intervals properly spaced to insure laboratory efficiency.

4.3.4 Fluid flow. The calibration and performance checks of the displaced fluid shall be determined by means of a calibrated flowmeter with a certified accuracy of 1 percent or better of reading.

4.3.5 Preliminary run-in and calibration. Any preliminary run-in of the stand as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the stand for testing. All instruments shall be calibrated and shall conform to the accuracies specified herein. A certified log of all preliminary running time, calibration data, et cetera, shall be submitted to the procuring activity.

4.3.6 Pump tests. The following tests shall be performed on one new high-pressure pump and one new low-pressure pump to establish the 1,000 hour life expectancy.

4.3.6.1 Efficiency tests. The high-pressure pump shall be connected to a calibrated power source and operated at 3,600 rpm to deliver 27 gpm flow at 3,000 psi while pumping hydraulic fluid conforming to MIL-H-5606 at 68° F with inlet pressure of 40 psig. Not more than 54 hp input shall be required to drive the pump.

4.3.6.2 Pump endurance. The high pressure pump shall be driven at 3.750 rpm for a minimum of 1,000 hours under the following conditions. Fluid temperatures during the 1,000 hours shall be maintained at 80° F, 25 percent of the time, 120° F 50 percent of the time, and 160° F the remaining 25 percent of the time for each flow condition. Inlet pressure shall be maintained at the minimum recommended by the pump manufacturer for 10 percent of the time, normal recommended pressure 80 percent of the time, and 300 psig 10 percent of the time for each flow condition. Output flow conditions shall be as follows:

Length of run (hours)	Output pressure (psig)	Output flow (gpm)
400	3,000	30
200	5,000	15
200	3,000	10
50	750	20
50	1,500	5
100	3,000	30

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4.3.6.2.1 Pump flow. By use of a quick-acting (0.1 second) solenoid valve in the pressure line, the pump shall be compensated from the preset flow to zero flow and vice versa once each minute during these tests. The pump case drain flow reading shall be recorded.

4.3.6.3 Compensation time. Inlet pressure, outlet pressure, and delivery of the pump shall be recorded at 15 gpm and 5,000 psi, 30 gpm at 3,000 psi, and 30 gpm at 1,000 psi to show compensating characteristics of the pump. The pump shall compensate from zero flow to full flow and vice versa in 0.05 second, and the pressure shall not exceed 133 percent of the selected pressure at any time.

4.3.6.4 Low pressure pump tests. The low pressure pump shall be tested at rated flow and speed for 1,000 hours under the following conditions:

Running time (hours)	Inlet conditions	Outlet pressure (psig)
300	125 psig	250
200	30 psig	155
300	15 inches Hg	95
100	10 inches Hg to 0 psig	110
100	125 psig	250

4.4 First article testing.

4.4.1 Test sample. One stand shall be subjected to the first article tests specified in 4.4.3.

4.4.2 Test report. After the first article tests are completed, three complete copies of a test report in accordance with MIL-STD-831 shall be supplied to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the first article test report (same quantity and type of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Description of the engineering reasoning verifying corrections made, to be made on production items or proposed to be made that substantiate the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

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e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. Estimates (± 5 percent of actual experienced) of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experiences of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

4.4.3 First article tests. The first article tests shall consist of all test described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests (see 4.5.1).

b. Sampling plan and tests (see 4.5.2).

4.5.1 Individual tests. Each stand shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1).

b. Flow test (see 4.6.2).

c. Cycling tests (see 4.6.3).

d. Fluid temperature test (see 4.6.4).

4.5.2 Sampling plan and tests.

4.5.2.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105. When MIL-STD-105 specifies an action by the Government, it shall at the option of the Government be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.5.2.2 Tests. Samples selected shall be subjected to the following tests as described under 4.6:

a. Semi-endurance test (see 4.6.5).

b. Examination (see 4.6.6).

4.6 Test methods.

4.6.1 Examination of product. The stand shall be inspected to determine compliance with requirements specified herein with respect to parts, accessibility, ease of maintenance, materials, workmanship, and markings.

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4.6.1.1 Leakage of valves. The filter bleed valve, fill system valve and high-pressure check valve shall be tested for leakage. The filter bleed valve leakage shall not exceed 2 drops in 5 minutes at a differential pressure of 275 psi; the fill system valve leakage shall not exceed 10 drops in 5 minutes at 150 psi differential pressure; and the high pressure check valve leakage shall not exceed 2 drops per minute at 5,000 psi differential pressure.

4.6.2 Flow tests. The pressure hose shall be connected through an external 1-inch size or larger high-pressure relief valve and back into the return hose. The reservoir selector valves shall be positioned in the stand reservoir position. The reservoir may be pressurized. The pump volume stop shall be set to the selected flow with the bypass valve full open. The zero flow pressure shall then be set with the bypass valve closed. The external relief valve shall be adjusted to obtain the selected flow and pressure. All data recorded shall be at the steady-state condition. Data shall be recorded for the following settings:

Volume stop setting (gpm)	Zero flow pressure setting (psi)	Approx full flow pressure (psi)
27	3,000	2,700
27	2,000	1,800
27	1,000	800
27	500	400
20	3,000	2,800
20	1,500	1,350
15	5,000	4,700
15	3,000	2,800
15	1,500	1,350
10	5,000	4,800
10	3,000	2,800
10	1,500	1,400
5	5,000	4,800
5	3,000	2,800
5	500	400

4.6.3 Cycling. For this test, a quick-acting (0.2 second or less) shutoff valve shall be installed between the pressure outlet connection and the relief valve specified in 4.6.2. The pressure and flow settings specified in 4.6.2 shall be repeated while cycling the shutoff valve to on and off positions at least 5 times at each setting. Steady-state pressure and flow readings shall be recorded for both the flow and no-flow conditions. The stand shall then be operated at 15 gpm flow and 5,000 psi pressure from each system for 1 hour and inspected for leakage of fluid.

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4.6.4 Fluid temperature. The temperature of the fluid entering the high pressure pump on the stand shall be raised above 160° F and the thermoswitch shall sound the warning horn or bell before the fluid temperature exceeds 165° F, but not before it reached 160° F. The warning horn or bell shall be audible in a radius of 25 feet from the stand in any direction when the stand is running at full load. On completion of this test, all filter elements in the unit shall be replaced with new elements.

4.6.5 Semi-endurance test. The stand shall be subjected to 20 percent of each test specified in 4.6.9 (a through e).

4.6.6 Examination. At the completion of the test specified in 4.6.5, each stand shall satisfactorily pass the tests specified in 4.6.2 and 4.6.3.

4.6.7 Initial disassembly inspection. The stand shall be disassembled to the maximum practical extent (see 6.4.3) and completely inspected, the breakdown weights determined, and measurements made which may be used to determine wear. In addition to manufacturer assembled items, the high pressure pump shall be disassembled. All gages, meters, and instruments shall be laboratory checked against certified calibrating equipment and failure of any item to meet the accuracy requirements specified shall be cause for rejection. All requirements which may be determined by weight or dimensional measurements and inspection shall be checked for compliance, and failure to meet these requirements shall be cause for rejection. Following this inspection, the stand shall be reassembled and subjected to the remainder of the tests.

4.6.8 Environmental tests. The stand shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-STD-810. Unless otherwise specified, upon completion of each test, the stand shall be subjected to the tests specified in 4.6.2 and 4.6.3. Failure of the stand to operate as required during or after each exposure, failure of any item or assembly due to deterioration from exposures in any test, or formation of any undesirable condition that cannot be remedied by routine maintenance shall be cause for rejection.

4.6.8.1 Low temperature. The stand shall be subjected to low temperature in accordance with method 502, procedure I, except the storage temperature shall be -65° F. The test circuit shall include MS28759-16-1200 pressure hose, relief valve to build up 3,000 psi pressure, and a MS28741-24-1200 return line. The entire test circuit shall be contained within the cold chamber, and the stand shall not be started until after the temperature of the stand and fluid have stabilized at -40° F. The fluid in the reservoir shall not be heated, prior to starting or during the test, from any external source. After starting, the flow and pressure shall be slowly increased to the rated output. The stand shall attain rated output within 30 minutes after starting and shall be operated at rated output for 10 minutes. The test required by 4.6.2 and 4.6.3 shall be performed at -40°F. All controls, gages, meters, valves, systems, and accessories shall be checked for satisfactory operation in the -40° F ambient temperature. Low temperature instructions as contained on the stand instrument panel shall be followed. The high pressure gage shall be accurate to within 1/2 percent of full scale. The system and fluid shall be stabilized at +20° F, and the output of the fill system checked. The fill system shall supply 2 gpm flow at 150 psig pressure with a fluid temperature of +20° F.

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4.6.8.2 High temperature. The stand shall be subjected to high temperature in accordance with method 501, procedure I. Upon completion of the exposure to high temperature, the ambient temperature shall be lowered to 125° F. The tests in 4.6.2 and 4.6.3 shall be performed at 125° F. An external cooler may be used to maintain the temperature of the fluid between 140° and 160° F. The high perssure gage shall be accurate to within ± 1/2 percent of full scale at all settings. The flowmeter shall be accurate within ± 1/2 gpm of actual flow. The system pump shall be adjusted to 27 gpm with the high pressure relief valve adjusted to 750 psig (routing full flow through the cooler). The unit shall continue to run at this flow and pressure setting until the fluid temperature stabilizes and remains stable for a minimum of 15 minutes. No external circuit cooler or reservoir shall be used for this portion of the test. The stand cooler shall maintain the fluid temperature below 160° F. After this test, the stand shall be inspected for leakage of fluid.

4.6.8.3 Fungus resistance. The stand shall be subjected to fungus in accordance with method 508, procedure I.

4.6.8.4 Sand and dust. The stand shall be subjected to sand and dust in accordance with method 510, procedure I.

4.6.8.5 Humidity. The stand shall be subjected to humidity in accordance with method 507, procedure I.

4.6.8.6 Salt fog. The stand shall be subjected to salt fog in accordance with method 509, procedure I.

4.6.8.7 Rain. The stand shall be subjected to rain in accordance with method 506, procedure I. The stand shall be positioned on the test chamber floor in its normal operating position. After exposure to the rain test for 1 hour and while still in the rain test, the stand shall be started and operated through the tests specified in 4.6.2 and 4.6.3. Failure of the stand to start or operate satisfactorily during this test shall be cause for rejection. Upon completion of this test, all surfaces that might collect moisture shall be inspected to determine adequacy of drainage facilities and any collection of water in excess of 1/8 inch deep or which could be detrimental to the performance of the stand shall be cause for rejection.

4.6.9 Endurance test. The stand shall be connected in series to a test rig through the 30-foot pressure hoses and 30-foot suction hoses specified herein. The test rig shall include a reservoir, restrictors to create the prescribed vacuum in the return lines, and a cooler or coolers to maintain fluid temperature at approximately 130° F. The ambient air temperature shall be maintained at 75° F or higher during the endurance test. Readings shall be recorded every hour of all the gages, meters, and instruments on the stand and test rig. Output flow, fluid temperatures, fluid pressure, inlet fluid temperature and pressure, motor voltage, amperage, power factor, and kilowatt shall be recorded each hour. The blower fan and cooler shall be operated continuously during all phases of the endurance and calibration test (see 4.6.12). Endurance test runs shorter than 1 hour shall not be counted as part of the endurance test. The stand shall first complete the test specified in 4.3.5 and shall then be subjected to the following 350 hours of endurance testing. The fill motor and pump and air compressor shall be operated continuously during this period. During the endurance tests, the flow shall be instantaneously (less than 0.1 second) varied

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from the present flow to zero flow, by means of a solenoid operated valve, and instantaneously returned to the preset flow at the end of 15 ± 2 seconds. This cycle shall be accomplished once each minute during the endurance test. Any damage to the pump, pump compensator, flow indicator, gearing, or other components shall be cause for rejection.

a. One hundred hours operation at 27 gpm and zero flow pressure setting at 3,000 psi. Pressure shall be discharged through the 1-inch outlets and returned through the 1-1/2 inch inlet with return pressure of 125 psig (reservoir bypassed).

b. Fifty hours operation at 15 gpm and zero flow pressure at 5,000 psi. Pressure shall be discharged through the 1-inch outlet and returned with pressure of 30 ±2 psig (reservoir in circuit).

c. One hundred hour operation at 27 gpm and zero flow setting at 3,000 psi with suction to the low-pressure pump reduced to 15 inches Hg vacuum. Pressure shall be discharged through the No. 1, 1-inch outlet and returned through the No. 2 1-1/2 inch inlet.

d. Fifty hours at 10 gpm with zero flow pressure setting at 1,000 psi and suction of 10 inches Hg to 0 psig.

e. Fifty hours operation at 10 gpm with zero flow pressure setting at 1,000 psi and return pressure of 125 psig.

4.6.9.1 Test data. The test specified in 4.6.2 shall be conducted. This data shall be compared with data recorded in 4.6.9. Results shall indicate that the stand is in satisfactory condition and that no failure or undue wear has resulted from the endurance test.

4.6.10 Electrical equipment. The electrical system shall be inspected to determine compliance with 3.9.2 and all items shall be shown to have underwriters' Laboratories approval or listing as required. During the last hour of the full capacity run specified in 4.6.9.a, the temperature rise in the motor windings shall be determined by the resistance method. If the temperature rise exceeds the safe temperature limits for the class insulation used, as prescribed by NEMA MG1, it shall be cause for rejection. The motors shall provide safe temperature rise at 6,000 feet pressure altitude and full load and 104° F ambient.

4.6.10.1 Thermoswitch operation. Operation of the thermoswitch shall be checked at least twice during the testing specified in 4.6.9, and shall sound the warning horn or bell when the fluid to the high pressure pump reaches not less than 160° F or not more than 165° F.

4.6.10.2 Pump motor. The pump motor shall be overloaded to 125 percent of full-load current and sustained until overload relays trip and shut off the motor. The overload relays shall trip before any damage is caused to the motor and shall not take longer than 15 mimutes to open the main contactor. Ambient temperature for motor and control shall be 104° ±9° F.

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4.6.10.3 Stand operation (27 gpm). The stand shall be operated for 2 hours at 27 gpm and zero flow pressure setting at 3,000 psi, with phase-to-phase voltage measured at the plug end of the electric cable reduced to 198 to 200V. The temperature rise in the motor windings shall be determined by the resistance method and shall not exceed the safe temperature limits for the class of insulation used. Ambient air temperature shall be held at 104° F, or higher, during this test.

4.6.10.4 Stand operation (440-V). The stand shall be reconnected for 440-V operation as specified in 3.9.2 and operated for 2 hours in 104° F, or over, ambient temperature. The temperature rise in the motor winding shall then be determined by the resistance method and shall not exceed the safe temperature limits for the class of insulation used.

4.6.10.5 Other motors. Other motors (fill pump and compressor) shall be tested and shown to meet the requirements of 4.6.10, 4.6.10.2, 4.6.10.3, and 4.6.10.4.

4.6.11 Air compressor and fill pump. The performance of the compressor and fill pump shall be checked at least once during each first article test. Failure of the drive couplings or any marked decrease in performance or failure of a component part of the pressurizing or fill system shall be cause for rejection. The output of the compressor and fill pump shall be determined and shall equal or exceed the requirements specified in 3.9.1.12.1 and 3.9.3.1.

4.6.12 Altitude. The contractor shall submit a test report with calculations showing that the stand will continuously deliver full-rated capacity while operating at a pressure altitude of 6,000 feet.

4.6.13 Radio interference. The stand shall be tested as a complete unit for radiated radio-interference suppression. The test shall be conducted in accordance with MIL-STD-826, except whenever a shielded room is not readily available, a test location shall be selected where the inherent background level and the area background level are within the limits established by MIL-STD-826 for class 1D equipment. The meter and test specimens shall have a common ground potential in substitution for the ground plane, and the antenna shall be limited to 6 feet from the perimeter of the test specimen. During the test, the stand shall be operating at 27 gpm and zero flow pressure setting of 3,000 psig.

4.6.14 Noise level. The noise level of the stand shall be measured at head level (5 feet 8 inches) in a circle of 25 feet radius centered on the geometric center of the unit at 12 positions equally spaced every 30° and at head level at the approximate operator position while the stand is operating at 27 gpm and 2,700 psi. The stand shall not exceed the noise levels shown below. Measurements shall be taken out-of-doors on a paved surface.

Cycles per second	Not exceeding (DB) Operator position	At 25 ft radius
37 to 75	106	91
75 to 150	96	81
150 to 300	88	73
300 to 600	85	70
600 to 1,200	84	69
1,200 to 2,400	83	68
2,400 to 4,800	82	67
4,800 to 9,600	81	66

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4.6.15 Valve tests. The flow control valve shall be tested by connecting the stand outlet to external flowmeter. With the pump set to operate at 1, 3, 5, 10, 15 gpm at 4,700 psi; 1, 3, 5, 10, 20, 27 gpm at 2,700 psi; and 1, 3, 5, 10, 20, 27 gpm at 1,000 psi, the flow control valve shall be operated to determine that it can hold the outlet flow steady at the desired value. Failure of the valve to regulate the flow to ± 1/4 gpm at any setting from 0 to 5 gpm and to ± 1 gpm at any setting from 5 to 27 gpm shall be cause for rejection. The flow control valve shall be tested for leakage with the pump operating at 3,000 psi. During these tests, the flow control valve settings shall not be influenced by vibration from the stand. Any leakage in excess of 1 drop per minute shall be cause for rejection.

4.6.16 Stand operation (titled position). The complete stand shall be rested on a platform and shall be kept from rolling by the parking brakes only. The platform supporting the stand shall be tilted in eight different directions, 45° apart, on an $8\text{-}1/2^{\circ}$ incline with the true horizontal plane. The stand shall be started and operated at rated output and 2,700 psi for 15 minutes in each position. The stand shall perform satisfactorily in every respect, and the parking brakes shall prevent the stand from moving on the platform. While tilted in each of the eight positions, 15 gallons of fluid shall be pumped from the reservoir and then replaced to check the capacity specified in 3.9.1.6.

4.6.17 Support test. A 200 pound weight shall be applied slowly and uniformly to an area 4-1/2 by 5 inches in at least six random positions on top of the stand housing, including a reading in the center of each panel. The stand shall not be permanently deformed by the application of the weight.

4.6.18 Mobility. The stand shall be subjected to and shall pass all tests specified in MIL-M-8090 for type II mobility. Upon completion of the mobility tests, the tests specified in 4.6.2 and 4.6.3 shall be repeated. Calibration of gages and minor adjustments will be permissible.

4.6.19 Serviceability. The stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular note shall be made with regard to provisions to prevent accumulation of dirt, snow, ice, et cetera, which may hinder servicing and operation.

4.6.20 Air transportability. The stand shall be subjected to the tests specified in MIL-A-8421. The stand shall be secured during this test by means of the lifting and tiedown fittings only. Upon completion of this test, the stand shall be inspected to determine adequacy of bracing, adequacy of tiedown and lifting fittings installation, and security of mounting of all components. The stand will not be required to be operable upon completion of the 8-g acceleration loading. If it can be shown by computation that the stand and all components will withstand the 8-g acceleration loading, the computation may be substituted in lieu of the test.

4.6.21 Backpressure indicator. The filter backpressure indicators shall be tested for proper operation. The indicators shall be tested with at least five different line pressures, including the highest and lowest operating pressure.

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4.6.22 Final disassembly inspection. The stand shall be disassembled to the maximum practical extent for a thorough examination of all parts to disclose failure, distortion, excessive wear, and other unsatisfactory conditions. Measurements made to determine wear shall be compared with similar measurements made during initial disassembly inspection. Stands not conforming to requirements specified herein shall be rejected.

4.6.23 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.24 Maintainability verification. Verification of the mainatainability requirements shall be in accordance with MIL-STD-471.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A, C, or Commercial as specified (see 6.2).

5.1.1 Level A. Each test stand shall be preserved in accordance with MIL-P-116, method I with P-2 preservative on all exposed bare metal surfaces. Hose shall be coiled and stored in the hose storage compartment. The ends of electrical cables shall be unwrapped with barrier material conforming to MIL-B-121, grade A, and secured in place with tape conforming to PPP-T-60.

5.1.1.1 Compressor. The compressor shall be preserved as specified in MIL-C-3600.

5.1.1.2 Motor. The motor shall be preserved as specified in MIL-E-16298.

5.1.2 Level C. The level C preservation for the test stand shall conform to the MIL-STD-2073 requirements for this level.

5.1.3 Commercial. The commercial preservation of the test stand shall be in accordance with the requirements of ASTM-D-3951.

5.2 Packing. Packaging shall be level A, B, C or Commercial, as specified (see 6.2).

5.2.1 Level A. Each test stand preserved as specified in 5.1.1 shall be packed in a MIL-C-104, type I, class 2, style A crate. The towbar shall be removed and attached to the base on the inside of the crate. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.2 Level B. Each test stand preserved as specified in 5.1.1 shall be shipped as a mobile (uncrated) unit.

5.2.3 Level C. The level C packing for the test stand shall conform to the MIL-STD-2073 requirements for this level.

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5.2.4 Commercial. The preserved test stand shall be packed in accordance with the requirements of ASTM-D-3951.

5.3 Marking.

5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM-D-3951.

5.4 General.

5.4.1 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.1.

6. NOTES

6.1 Intended use. The type MK-1 stand is intended for use in ground testing and maintenance of aircraft hydraulic systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. That the contractor submit proof that the electrical system specified in 3.9.2 which will be supplied under this specification meets the requirements of NFPA 70. The label or listing of the Underwriters' Laboratories, Incorporated, will be accepted as evidence of conforming to the specification requirements. In lieu of the label or listing, the contractor may submit independent proof satisfactory to the procuring activity that the electrical system conforms to the requirements of NFPA 70. Compliance with these requirements does not absolve the contractor from complete compoiance with the other requirements of this specification.

c. Point of inspection

d. Applicable levels of preservation, packaging and industrial/commercial packaging (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior

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Government approval is presently appropriate for the pending contract.

6.4 Definitions. For the purpose of this specification, the following definition will apply:

6.4.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of non-failure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent.

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = \exp(-X) \text{ where } X = \frac{\text{Mission time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the poisson/chi squared distribution, assuming an exponential (2° freedom) distribution even though the failure rate for the test time is zero.

6.4.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.4.3 Disassembled to the maximum practical extent. Disassembled to the maximum practical extent is defined as disassembly of all moving contacting surfaces which have been assembled by the manufacturer of the stand and which are subject to wear (gears, bearings, control mechanisms, et cetera).

6.5 Pump mountings. The high pressure pumps on this stand are mounted on standard accessory drive pads. Aircraft pumps can be substituted by the use of the following parts:

AN4139	Adapter-type XVI drive flange.
AN4182	Adapter-type XVII drive spline.

6.6 Pump rating and testing. The high-pressure pumps are rated (see 3.9.1.1) and tested at 3,750 rpm (see 4.3.6) at 30 gpm, 3,000 psi to insure interchangeability with pumps on gasoline-engine-driven stands.

6.7 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ASCC AIR STD 11/14D. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

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6.8 Subject term (key word) listing.

Stand, Test, Aircraft Hydraulic System, Electronic Motor Driven, Type MK-1

Test Stand, Aircraft Hydraulic System, Electric Motor Driven, Type MK-1

Tester, Aircraft Hydraulic System, Electric Motor Driven, Type MK-1

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AV
Air Force - 99

Preparing activity:
Air Force - 82

(Project 4920-0411)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-26000	Stand, Test, Aircraft Hydraulic System, Electric Motor Driven, Type MK-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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